

RESEARCH ARTICLE

Habitat conditions filter stronger for functional traits than for phenology in herbaceous species

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Abstract

An increasing number of studies in botanical gardens are investigating species' responses to climate change. However, the influence of local environmental or habitat conditions such as soil nutrient status or microclimate on phenology and the link between morpho-physiological functional traits and phenological stages are poorly understood, making it difficult to extrapolate patterns from botanical gardens to natural environments. Therefore, we selected herbaceous species growing in two semi-natural habitats, namely, semi-dry grasslands (SDGs) and mesophilic grasslands (MGs) and the botanical garden of Jena (Germany) to investigate the influence of habitat conditions on interspecific and intraspecific patterns in phenology, functional traits and their associations. For 16 species, we monitored leaf and flowering phenology weekly for 133 populations from the three habitats, measured morpho-physiological traits (i.e., whole plant, leaf and reproductive traits), as well as habitat conditions and compared the measurements across habitats. Multivariate analyses revealed that morpho-physiological traits conspicuously showed stronger differences between habitats compared to phenological traits. Populations on MG showed temporal niche segregation, whereas populations on SDG showed flowering synchrony. Boosted Regression Trees showed that morpho-physiological traits, especially reproductive traits, strongly influenced phenological traits and that the trait-phenology relationships were highly habitat-specific. We conclude that species phenology is broadly similar between botanical gardens and local habitats. However, phenological responses to the environment may be constrained by a certain suite of correlated traits due to ecological plant strategies that vary across habitats. The effect of habitat conditions on morpho-physiological functional traits and phenology-trait relationships is important and should not be neglected at local scales, implying consequences at larger scales.

KEYWORDS

botanical garden, first flowering day, grassland, habitat conditions, phenology, plant functional traits

TAXONOMY CLASSIFICATION

Functional ecology

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